

## *Principle Component Analysis (PCA)*

*Step – 1: Read the data*

*Step – 2: Preprocess the data(EDA)*

*Step – 3: Model development*

*Step – 4: Model evaluation*

*Step – 5: Model Tuning (Hyper parameter tuning)*

*Step – 6: Model deployment*

### *Under data analysis*

- *Feature engineering*
  - *Encoding*
  - *Scaling*
- *Feature selection*
  - *Correlation input vs output*
  - *VIF*
  - *p value*
  - *forward selection vs backward selection*
  - *Entropy*

*Overfitting – Underfitting – Normal fitting*

*Bias – variance*

*train error – test error*

*Performance metrics*

*Important points to be considered always:*

- 1) *We always concern about features*
- 2) *Good features we have, then model will be good*
- 3) *Overfitting happens : Model complexity increases*
- 4) *Model complexity means*
  - Linear regression: More number of features*
  - Decision Tree : More Depth*
- 5) *A data has 100 columns: 100 Dimensions of data*
- 6) *Regularization methods*
- 7) *lasso method can use as feature selection*
- 8) *Multicollinearity should not be there between the input column*
- 9) *Input columns should be independent each other*

**A) only Important features should consider**

**B) Features should be independent each other: No correlation between the features**

*Loss of the data: When you select some feature we might lose the data*

*Requirement:*

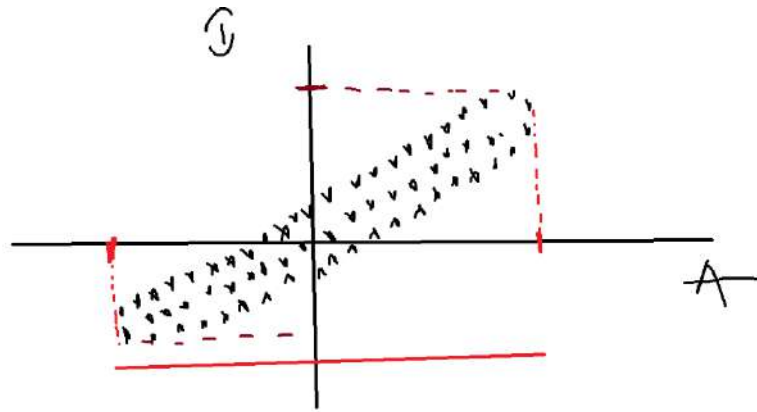
- 1) *We should reduce the features*
- 2) *We should not lose the data*
- 3) *The features should not have multicollinearity (features should be independent)*

*Answer: PCA*

### **PCA Process:**

*If a data has more dimensions(features), The processing time very much increases  
we required more infrastructure  
We require more time  
This is called as **Curse of Dimensionality***

*PCA also called as Dimension reduction technique*



### Rule – 1:

*If anyone asks choose one variable between Age and income*

*It is very tough to select one variable between Age and Income*

*Because The data varies along Age and along Income almost same and huge*

$$\text{var}(\text{Age}) = \text{var}(\text{Income})$$

### Rule – 2:

*In order to choose any column, the data varies along that column should be very very high*

$$\text{var}(\text{Age}) \gg \gg \gg \text{Var}(\text{Income}) \cong \text{Age column}$$

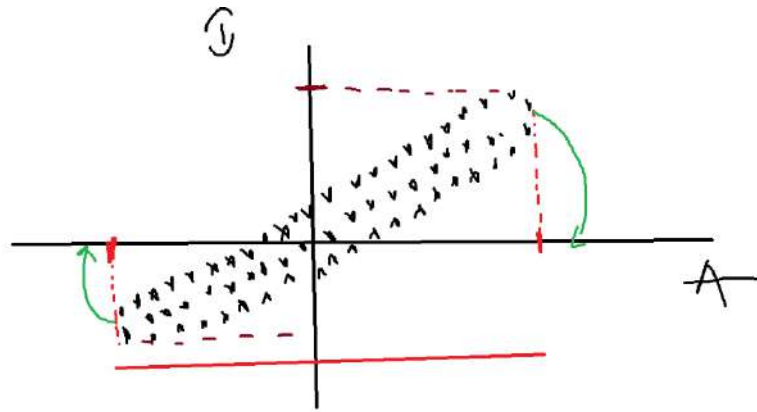
$$\text{var}(\text{Income}) \gg \gg \gg \text{Var}(\text{Age}) \cong \text{Income column}$$

$$\text{var}(\text{Age})(98\%) \gg \gg \gg \text{Var}(\text{Income})(2\%) \cong \text{Age column}$$

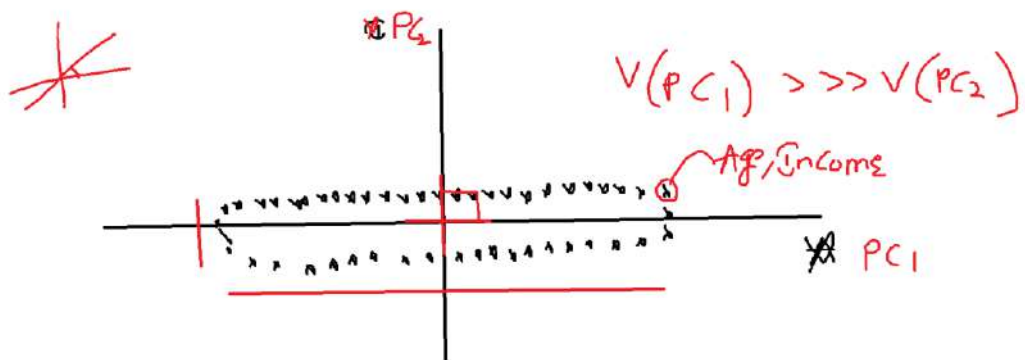
*The data along Age has 98%, The data along Income has 2%*

*It is okay we can have 98% of data enough to develop the model*

- Now we will project the data points on new axis
- This data is called as Transformed data
- these 98% of data is a transformed data, having both age and income information
- 2% of data also transformed along another axis having both age and income information
- both axis has the data, both axis has income and age data, but only one axis data
- variance is more, so we are choosing that axis data
- The data can be on transform on two axis
- These axis called as PC's



Observations are:



- we have original dimension space with Age and column

- The variance along two columns are equal
- Both are having correlation
- Now we are creating a new dimension space
  - The new dimensions space is also having two axis
  - $PC_1$  : Principle component – 1
  - $PC_2$ : Principle component – 2
  - The new Dimension space has different variances
  - The data varies along  $PC_1$  is more than  $PC_2$
  - $Var(PC_1) \gg Var(PC_2)$
  - Both  $PC_1$  and  $PC_2$  are independent each other
  - Orthogonal each other
  - Perpendicular each other
  - Having  $90^\circ$  phase shift
  - $PC_1 = vector * Age + vector * Income$
  - $PC_2 = vector * Age + vector * Income$

For example originally we have 10 columns

10 PC :  $PC_1, PC_2, PC_3, \dots, PC_{10}$

$var(PC_1) > var(PC_2) > \dots > var(PC_{10})$

$var(PC_1) + var(PC_2) + \dots Var(PC_{10}) = 100\%$

$Var(PC_1) = 95\% + var(PC_2) = 3\% + var(PC_3) = 1\% + \dots = 100\%$

$PC_1 = w_{11} * X_1 + w_{12} * X_2 + \dots + w_{110} * X_{10}$

$PC_2 = w_{21} * X_1 + w_{22} * X_2 + \dots + w_{210} * X_{10}$

$PC_3 = w_{31} * X_1 + w_{32} * X_2 + \dots + w_{310} * X_{10}$

$PC_1 = w_{11} * X_1 + w_{12} * X_2 + \dots + w_{110} * X_{10}$

- Number of PCs = Number of Original columns
- Variance of each PC is different , In above example first 3PCs only covering almost 98% of data
- In order to train the model 98% of data is enough
- So Instead of 10Pcs , we can choose only 3Pcs
- Means instead of 10 Columns (Dimensions) we can choose only 3 Columns(Dimensions)
- This is the way we are reducing the Dimensions
- Each PC data = Linear combination of Original data
- So we are not losing the data
- The New dimension space PC's all are orthogonal to each other

1) When you want to create a New Dimension space

2) In which direction the data should project

○ The data is projecting any direction =====> Axis

3) How much you want project

○ More projection ===== Variance in that direction is more

The direction gives by : Eigen vectors

The amount of projection given by: Eigen values

We need to calculate the Eigen values and Eigen vectors of Covariance Matrix

Eigen vectors wil give the direction

Eigen values will give the Variance

$$Ax = \lambda x$$

$$A = \text{matrix}$$

$$x = \text{eigen vector}$$

$$\lambda = \text{eigen value}$$

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

$$Ax = \lambda x$$

$$Ax - \lambda x = x(A - \lambda I) = 0$$

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = 0$$

$$\left| \begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix} \right| = 0$$

$$(1 - \lambda) * (4 - \lambda) - 10 = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 - 10 = 0$$

$$\lambda^2 - 5\lambda - 6 = 0$$

$$\lambda^2 - 6\lambda + 1\lambda - 6 = 0$$

$$\lambda(\lambda - 6) + 1(\lambda - 6) = 0$$

$$(\lambda - 6)(\lambda + 1) = 0$$

$$\lambda = 6 \text{ and } \lambda = -1$$

*Matrix rank = 2x2 : Number of Eigen values = 2, Eigen vectors = 2*

$$x(A - \lambda I) = 0$$

$$\lambda = 6$$

$$\left( \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} - 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) * \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\left( \begin{bmatrix} 1-6 & 2 \\ 5 & 4-6 \end{bmatrix} \right) * \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\left( \begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \right) * \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\lambda = -1$$

$$\left( \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} + 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) * \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\left( \begin{bmatrix} 1+1 & 2 \\ 5 & 4+1 \end{bmatrix} \right) * \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\left( \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \right) * \begin{bmatrix} a \\ b \end{bmatrix} = 0$$

$$\text{The first Eigen vector} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$\text{The first Eigen value} = 6$$

$$\text{The second Eigen vector} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{The second Eigen value} = -1$$

Original matrix :  $A = \text{Covariance matrix}$

Data spread ness in both Directions

From there we are creating two more directions (Two vectors)

- Eigen vector – 1:  $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$
- Eigen vector – 2:  $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$
- Amount of projection is :
  - The first Eigen value = 6 units long the First eigen vectors
  - The Second Eigen value =  $-1$  units long the second eigen vectors
- Original data projected in two seperated directions
- The direction becomes axis
- That axis are  $Pc_1$  and  $Pc_2$
- $PC_1$  has more projection means more variance : 6 units
- $PC_2$  has less projection means less variance:  $-1$  units

**New data = Eigenvectors \* old data**

$$Pc_1 = \text{Age} * \text{Eigen vector}_1 + \text{Income} * \text{Eigen vector}_1$$

$$Pc_2 = \text{Age} * \text{Eigen vector}_2 + \text{Income} * \text{Eigen vector}_2$$

$$\text{var}(Pc_1) \gg \gg \text{var}(pc2)$$

$$\text{cov}(Pc_1, Pc_2) \cong 0$$

1) when a data point is changing

towards to another direction

means data value change

$$\text{New data} = \text{olddata} * \text{vector}$$

2) Vectors will say to my data points

A) Go in that direction

B) Go some amount of duration

3) This duration is nothing but data travel

along that direction

direction means new axis

travel means=variance

data = olddata\*vector